⊛ On a la matrice de transition sous forme que:

$$T^d = (2\pi)^{-3} \int e^{i\Delta r_0} A_{mo}(r_0)\, dr_0 \quad \text{avec} \quad A_{mo} = \langle m | \frac{1}{r_{01}} - \frac{1}{r_0} | 0 \rangle$$

Cas élastique: $1S \to 1S \Rightarrow A_{oo} = \langle 0 | \frac{1}{r_{01}} - \frac{1}{r_0} | 0 \rangle$

⊛ A_{oo} c'est une moyenne du potentiel statique, ce potentiel s'écrit comme:

$$V_{st}(r_0) = \langle 0 | \frac{1}{r_{01}} - \frac{1}{r_0} | 0 \rangle \quad \text{avec} \quad |0\rangle = \frac{e^{-r_1}}{\sqrt{\pi}}$$

$$\Rightarrow V_{st}(r_0) = \frac{1}{\pi} \int e^{-2r_1} \left(\frac{1}{r_{01}} - \frac{1}{r_0} \right) dr_1 \quad (dr_1 = r_1^2 dr_1 d\Omega_1)$$

* $\frac{1}{r_{01}}$ peut être développée en harmonique sphérique tel que:

$$\frac{1}{r_{01}} = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{4\pi}{2l+1} \frac{r_<^l}{r_>^{l+1}} Y_{lm}(\hat{r}_1) Y_{lm}^*(\hat{r}_0)$$

$$\Rightarrow V_{st}(r_0) = \frac{1}{\pi} \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{4\pi}{2l+1} Y_{lm}^*(\hat{r}_0) \int d\Omega_1 Y_{lm}(\hat{r}_1) \int_0^{+\infty} r_1^2 dr_1 e^{-2r_1} \frac{r_<^l}{r_>^{l+1}}$$
$$- \frac{1}{\pi} \int d\Omega_1 \int_0^{+\infty} e^{-2r_1} r_1^2 dr_1 \cdot \frac{1}{r_0}$$

$$\Rightarrow V_{st}(r_0) = 4 \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{Y_{lm}^*(\hat{r}_0)}{2l+1} \sqrt{4\pi}\, \delta_{l0} \delta_{m0} \int_0^{+\infty} r_1^2 dr_1 e^{-2r_1} \frac{r_<^l}{r_>^{l+1}}$$
$$- \frac{1}{\pi} 4\pi \int_0^{+\infty} dr_1\, r_1^2 e^{-2r_1} \frac{1}{r_0}$$

$Y_{oo} = Y_{oo}^* = \frac{1}{\sqrt{4\pi}}$

$\int Y_{lm} Y_{l'm'}^* dr = \delta_{ll'} \delta_{mm'}$

$\int d\Omega_1 = 4\pi$

⊛ On remplace $l=0$ et $m=0$, on calcule les deux sommes et on trouve:

$$V_{st}(r_0) = 4 \int_0^{+\infty} r_1^2 dr_1 e^{-2r_1} \frac{1}{r_>} - 4 \int_0^{+\infty} r_1^2 dr_1 e^{-2r_1} \frac{1}{r_0}$$

$$V_{st}(r_0) = 4 \int_0^{+\infty} dr_1\, r_1^2 e^{-2r_1} \left(\frac{1}{r_>} - \frac{1}{r_0} \right)$$

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$$V_{st} = 4\int_{r_0}^{+\infty} dr_1\, r_1^2 e^{-2r_1}\left[\frac{1}{r_1} - \frac{1}{r_0}\right]$$

$$\int_0^{+\infty} f(r_0, r_1)\, dr_1 = \underbrace{\int f(r_1 \to r_0) dr_1}_{0} + \int f(r_1 \to r_1) dr$$

$$\Rightarrow V_{st}(r_0) = 4\left[\int_{r_0}^{+\infty} r_1\, dr_1\, e^{-2r_1} - \frac{1}{r_0}\int_{r_0}^{+\infty} r_1^2\, dr_1\, e^{-2r_1}\right]$$

* On utilise le changement de variable : $x = r_1 - r_0$

$$\Rightarrow V_{st}(r_0) = 4\left[\int_0^{+\infty} dx\,(x+r_0)\, e^{-2(x+r_0)} - \frac{1}{r_0}\int_0^{+\infty} (x+r_0)^2\, dx\, e^{-2(x+r_0)}\right]$$

$$\Rightarrow V_{st}(r_0) = 4\left[e^{-2r_0}\left(\int_0^{+\infty} x\, dx\, e^{-2x} + r_0\int_0^{+\infty} dx\, e^{-2x}\right) - \frac{e^{-2r_0}}{r_0}\left(\int_0^{+\infty} x^2 dx\, e^{-2x} + \int_0^{+\infty} 2x r_0\, dx\, e^{-2x} + \int_0^{+\infty} r_0^2 e^{-2x} dx\right)\right]$$

$$\Rightarrow V_{st}(r_0) = 4e^{-2r_0}\left(\frac{\Gamma(2)}{2^2} + r_0\frac{\Gamma(1)}{2^1} - \frac{\Gamma(3)}{r_0 2^3} + \frac{2r_0\Gamma(2)}{2^2 r_0} + \frac{r_0^2}{r_0}\frac{\Gamma(1)}{2^1}\right)$$

$$\int_0^{+\infty} x^{m-1} e^{-yx} dx = \frac{\Gamma(m)}{y^m} = \frac{(m-1)!}{y^m}$$

$$\Rightarrow V_{st}(r_0) = 4e^{-2r_0}\left(\frac{1}{4} + \frac{r_0}{2} - \frac{1}{4r_0} - \frac{1}{2} - \frac{r_0}{2}\right)$$

$$\boxed{V_{st}(r_0) = -\left(1 + \frac{1}{r_0}\right)e^{-2r_0}} \quad \text{finalement}$$

⊛ On remplace $V_{st}(r_0)$ dans la matrice de transito :

$$T_d = -(2\pi)^3\int_0^{+\infty} e^{i\Delta r_0}\left(1 + \frac{1}{r_0}\right)e^{-2r_0}\, dr_0$$

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$e^{i\Delta r_0}$ est une harmonique sphérique.

J_l : fct de Bessel

$J_0(\Delta r_0) = \frac{\sin \Delta r_0}{\Delta r_0}$

$$e^{i\Delta r_0} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^{l} i^l J_l(\Delta r_0) Y_{lm}(\hat{r}_0) Y_{lm}^*(\hat{\Delta})$$

$$\Rightarrow T_d = -(2\pi)^{-3} \int_0^{+\infty} 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^{l} i^l J_l(\Delta r_0) Y_{lm}(\hat{r}_0) Y_{lm}^*(\hat{\Delta}) \times \left(1 + \frac{1}{r_0}\right) e^{-2r_0} dr_0$$

$$\Rightarrow T_d = -4\pi (2\pi)^{-3} \int_0^{\infty} \sum_{l=0}^{\infty} \sum_{m=-l}^{l} i^l J_l(\Delta r_0) Y_{lm}^*(\hat{\Delta}) \sqrt{4\pi} \int d\Omega Y_{lm} Y_{00} \times \left(1 + \frac{1}{r_0}\right) e^{-2r_0} r_0^2 dr_0$$

$$\Rightarrow T_d = -4\pi (2\pi)^3 \int_0^{\infty} \sum_{l=0}^{\infty} \sum_{m=-l}^{l} i^l J_l(\Delta r_0) Y_{lm}^*(\hat{\Delta}) \sqrt{4\pi}\, \delta_{l0} \delta_{m0} \times \left(1 + \frac{1}{r_0}\right) e^{-2r_0} r_0^2 dr_0$$

On remplace $l=0$, $m=0$

$$\Rightarrow T_d = -4\pi (2\pi)^{-3} \int_0^{\infty} J_0(\Delta r_0)\, r_0^2\, dr_0\, e^{-2r_0} \left(1 + \frac{1}{r_0}\right)$$

$$= \underbrace{-4\pi (2\pi)^{-3} \int_0^{+\infty} J_0(\Delta r_0)\, r_0^2 dr_0 e^{-2r_0}}_{A} \underbrace{- 4\pi (2\pi)^{-3} \int_0^{+\infty} J_0(\Delta r_0) r_0^2 dr_0 \frac{e^{-2r_0}}{r_0}}_{B}$$

$$= A + B$$

$\sin \Delta r_0 = \frac{e^{i\Delta r_0} - e^{-i\Delta r_0}}{2i}$

$\int_0^{+\infty} x^n e^{-\mu x} dx = \frac{n!}{\mu^{n+1}}$

⊛ On calcule A et B :

$$A = -4\pi (2\pi)^{-3} \int_0^{+\infty} \frac{\sin(\Delta r_0)}{\Delta} r_0\, dr_0\, e^{-2r_0}$$

$$= \frac{-4\pi (2\pi)^{-3}}{2i\Delta} \int_0^{+\infty} \left(e^{i\Delta r_0} - e^{-i\Delta r_0}\right) r_0 e^{-2r_0} dr_0$$

$$= \frac{-1}{4\pi^2 i\Delta} \left(\int_0^{+\infty} r_0 e^{-(2-i\Delta) r_0} dr_0 - \int_0^{+\infty} e^{-(2+i\Delta) r_0} dr_0 \right)$$

$$= \frac{-1}{4\pi^2 i\Delta} \left(\frac{1!}{(2-i\Delta)^2} - \frac{1!}{(2+i\Delta)^2} \right) = \frac{-1}{4\pi^2 i\Delta} \left(\frac{4 + 4i\Delta - \Delta^2 - (4 - 4i\Delta - \Delta^2)}{(4+\Delta^2)^2} \right)$$

$$\Rightarrow \boxed{A = \frac{-2}{\pi^2 (4+\Delta^2)^2}}$$

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$$B = \frac{-4\pi}{(2\pi)^3} \int_0^{+\infty} dr_0\, r_0\, e^{-2r_0} J_0(\Delta r_0)$$

$$\int_0^{+\infty} e^{-px}\, dx = \frac{1}{p}$$

$$B = \frac{-1}{4\pi^2 i\Delta} \int_0^{+\infty} \left(e^{i\Delta r_0} + e^{-i\Delta r_0}\right) e^{-2r_0}\, dr_0$$

$$B = \frac{-1}{4\pi^2 i\Delta} \left(\int_0^{+\infty} e^{-(2-i\Delta)r_0}\, dr_0 - \int_0^{+\infty} e^{-(2+i\Delta)r_0}\, dr_0 \right)$$

$$B = \frac{-1}{4\pi^2 i\Delta} \left(\frac{1}{2-i\Delta} - \frac{1}{2+i\Delta} \right) = \frac{-1}{4\pi^2 i\Delta} \left(\frac{2+i\Delta-2+i\Delta}{4+\Delta^2} \right)$$

$$\Rightarrow B = \frac{-1}{2\pi^2} \frac{1}{4+\Delta^2} \Rightarrow \boxed{B = -\frac{4+\Delta^2}{2\pi^2(4+\Delta^2)^2}}$$

Alors T_d obtient

$$T_d = A + B = \frac{-2}{\pi^2(4+\Delta^2)^2} - \frac{4+\Delta^2}{2\pi^2(4+\Delta^2)^2}$$

$$T_d = \frac{-4-4-\Delta^2}{2\pi^2(4+\Delta^2)^2} \Rightarrow \boxed{T_d = -(2\pi^2)^{-1} \frac{8+\Delta^2}{(4+\Delta^2)^2}}$$

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